Server Access logs and Storage Classes.

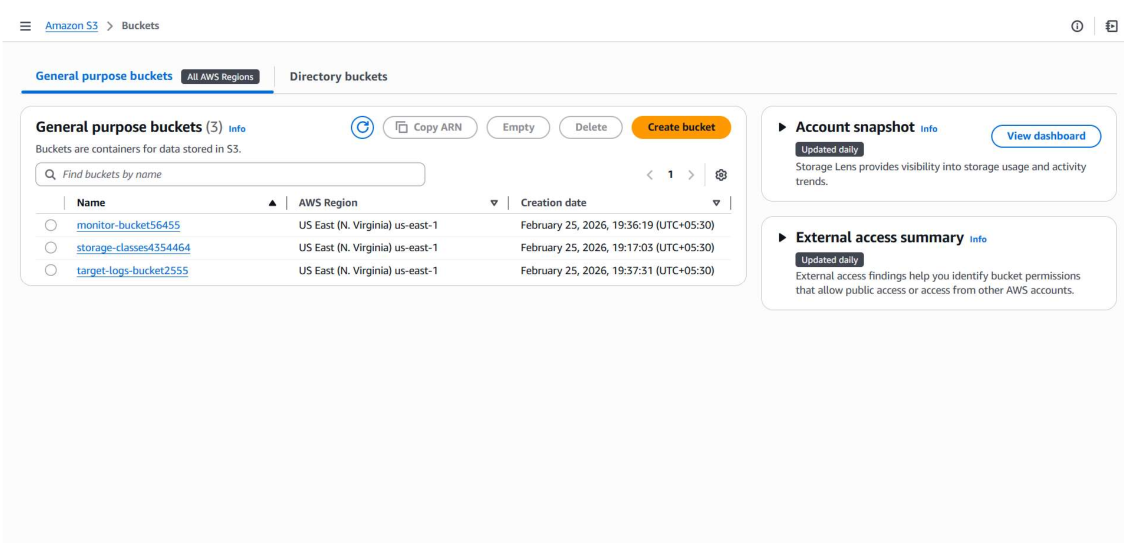
Server Access Logs in S3

In Amazon S3, **Server Access Logging** records detailed requests made to your bucket.

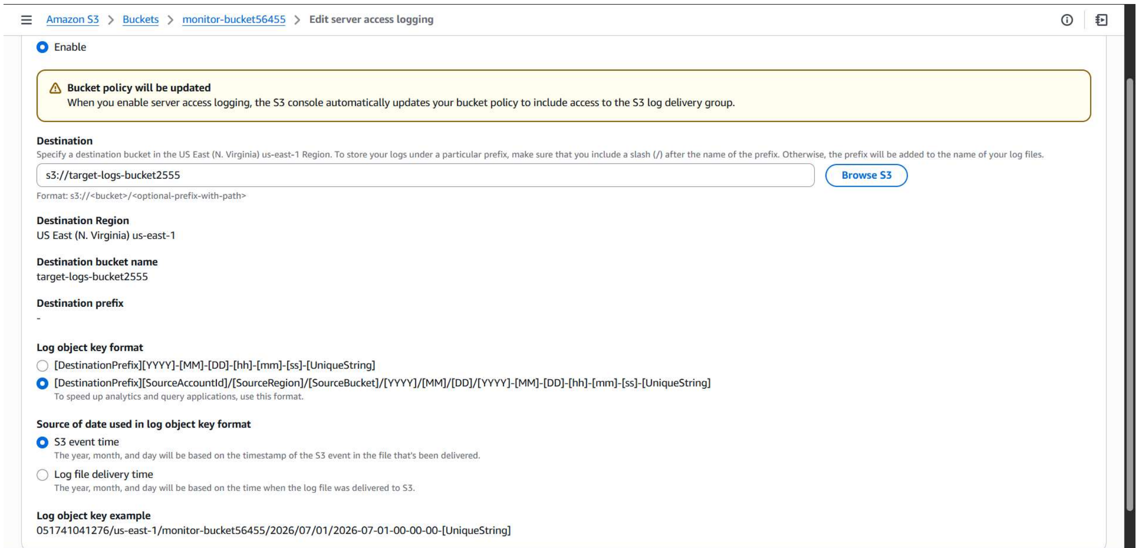
That means:

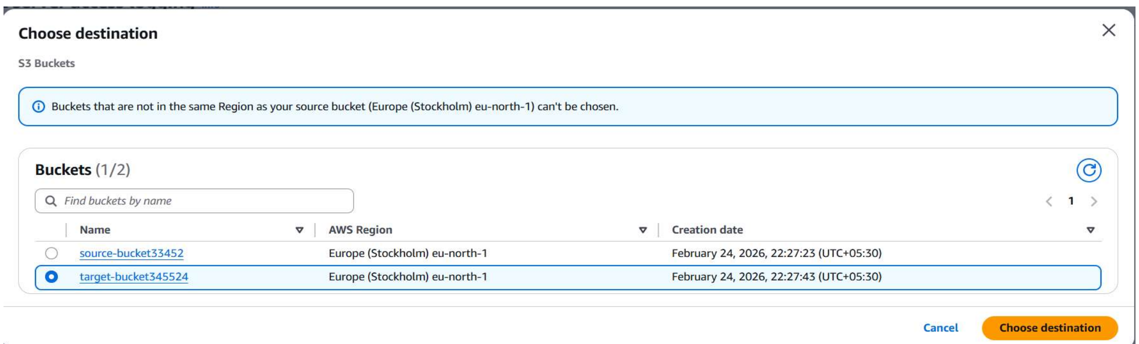
- Who accessed your bucket
- What object was accessed
- When it happened
- From which IP
- Whether it succeeded or failed

I have created two new buckets for the Server Access Logs.

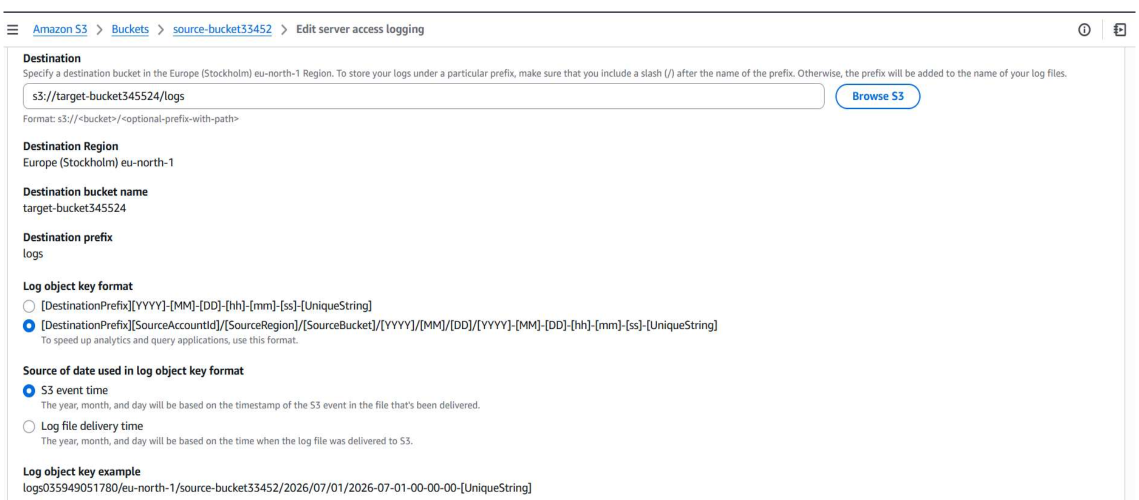


One is target and the other one is source. In the properties tab of the source bucket we have to link the log file to the target bucket (works when both the buckets in same region).

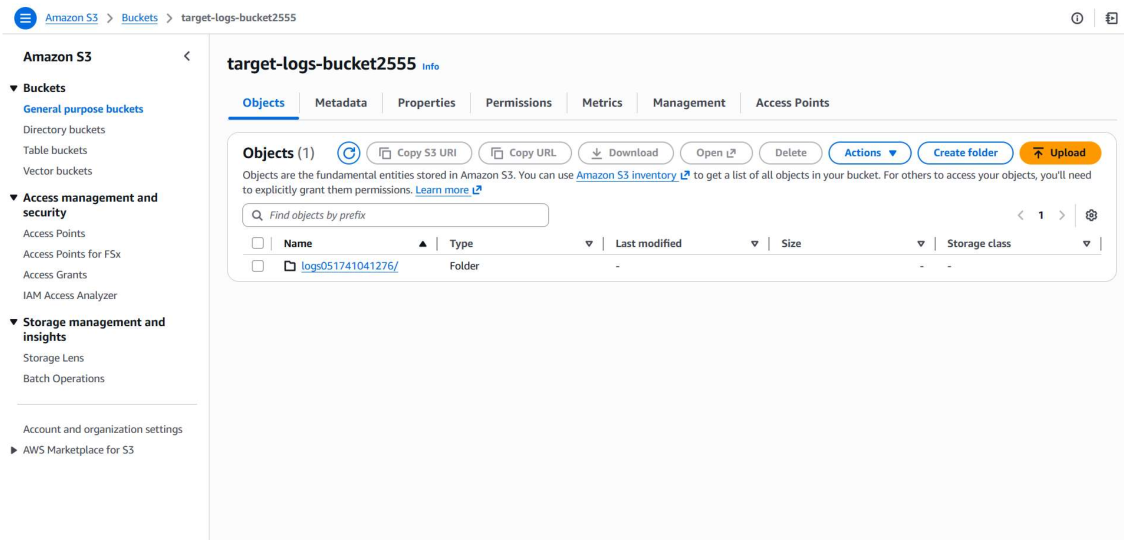


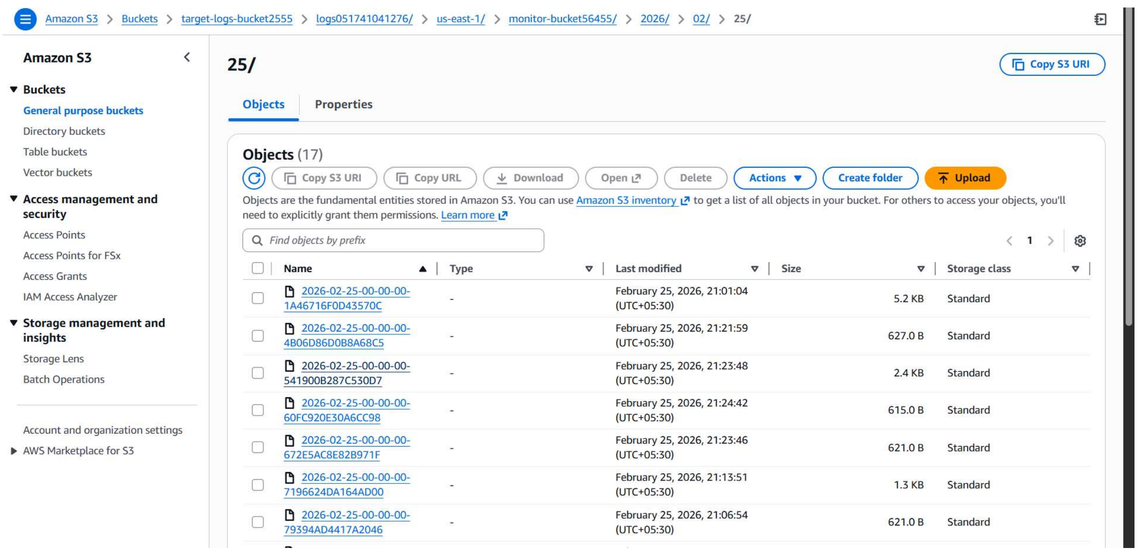


We have to specify which format we want for logs



Then we have to wait for about 15 to 30 mins while bucket creating the log file.





Storage Classes in Amazon S3 (Theory)

In Amazon S3, a **storage class** defines the cost, availability, durability, and retrieval characteristics of stored objects.

All S3 storage classes provide **11 nines durability (99.999999999%)**, but they differ in:

- Storage cost
- Retrieval cost
- Data availability
- Access latency
- Minimum storage duration

The selection of storage class depends on how frequently the data is accessed and how quickly it must be retrieved.

1️⃣ S3 Standard

S3 Standard is designed for **frequently accessed data**.

Characteristics:

- High availability (99.99%)
- Low latency
- Multi-Availability Zone storage
- No retrieval fee

Use Cases:

- Web applications
- Mobile applications
- Real-time analytics
- Frequently accessed APIs

It is the default storage class in S3.

2️⃣ S3 Intelligent-Tiering

S3 Intelligent-Tiering automatically moves objects between different access tiers based on usage patterns.

Characteristics:

- Automatic cost optimization
- No retrieval charges for frequent and infrequent tiers
- Small monitoring fee per object
- Designed for unpredictable access patterns

Use Cases:

- Applications with unknown usage behavior
- Data lakes

- Large datasets with variable access frequency

This class removes the need for manual lifecycle management.

3️⃣ S3 Standard-Infrequent Access (Standard-IA)

S3 Standard-IA is designed for data that is accessed less frequently but requires rapid access when needed.

Characteristics:

- Lower storage cost than Standard
- Retrieval fee applies
- Multi-AZ storage
- Minimum storage duration: 30 days

Use Cases:

- Backup files
- Disaster recovery data
- Monthly or quarterly reports

It balances cost savings with high availability.

4️⃣ S3 One Zone-Infrequent Access (One Zone-IA)

S3 One Zone-IA stores data in a single Availability Zone.

Characteristics:

- Lower cost than Standard-IA
- Retrieval fee applies
- Data stored in one AZ only
- Minimum storage duration: 30 days

Use Cases:

- Secondary backups
- Re-creatable data
- Test or development data

It offers reduced cost but lower resilience compared to multi-AZ classes.

5️⃣ S3 Glacier Storage Classes

Glacier storage classes are designed for archival and long-term data retention.

a) Glacier Instant Retrieval

- Millisecond access
- Lower cost than Standard-IA
- Minimum storage duration: 90 days

b) Glacier Flexible Retrieval (formerly Glacier)

- Retrieval time: Minutes to hours
- Very low storage cost
- Minimum storage duration: 90 days

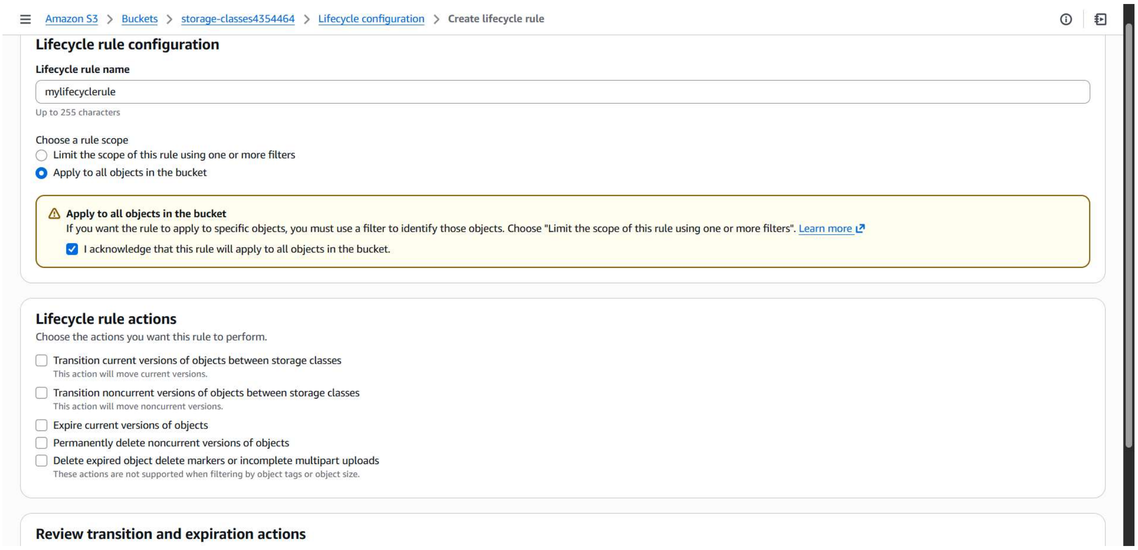
c) Glacier Deep Archive

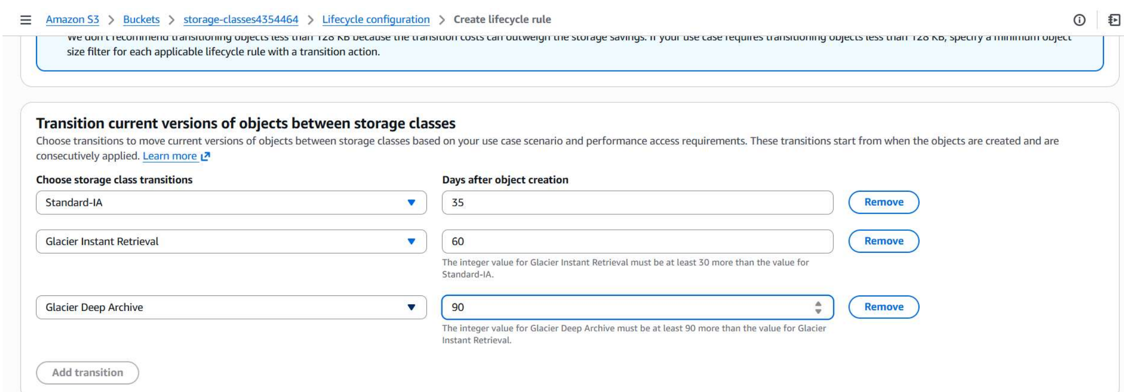
- Retrieval time: 12–48 hours
- Lowest storage cost
- Minimum storage duration: 180 days

Use Cases:

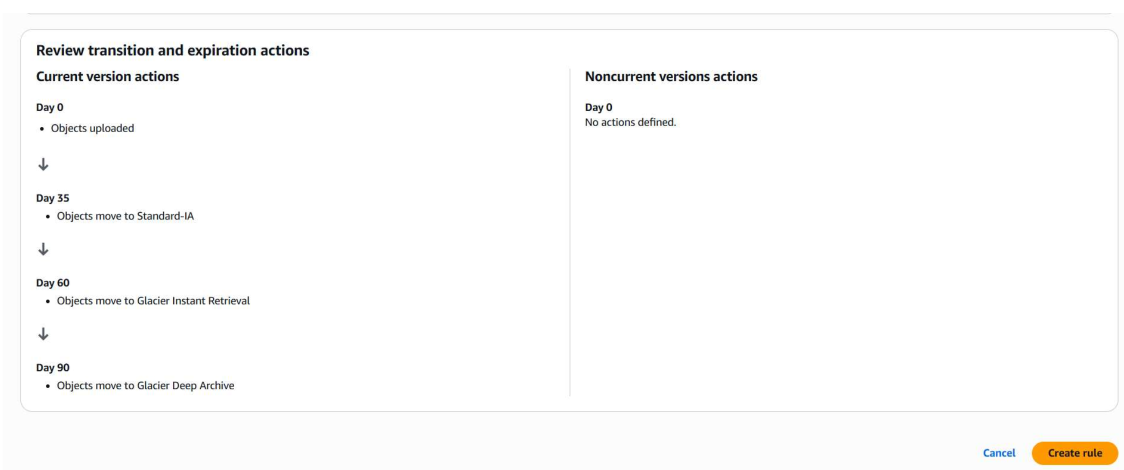
- Compliance data
- Long-term archives
- Legal or regulatory storage

In the created bucket we have to change the Storage Classes in the Management option using the Lifecycle Configuration.

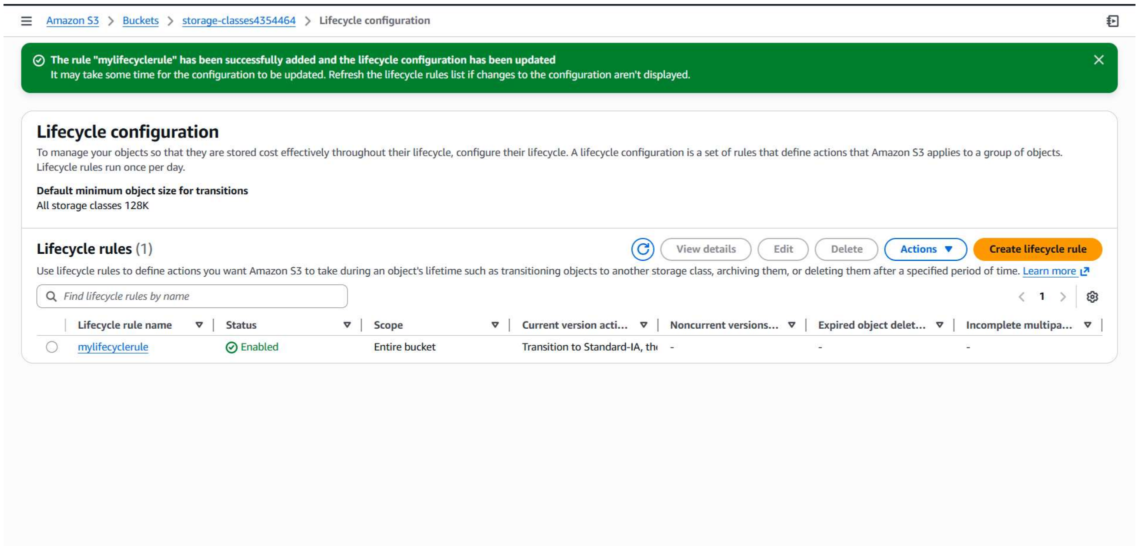




These are the review of the Storage class you have been creating.



Above are the steps to create the Storage Classes.



The rule have been created successfully.